

# KRUTARTH ASHAR

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## OBJECTIVE

Artificial Intelligence and Data Science engineering student with hands-on experience in Python, SQL, data analysis, machine learning, ETL pipelines, and data visualization. Experienced in transforming raw and unstructured data into clean, traceable datasets, performing exploratory analysis, and evaluating data-driven models. Seeking a Data Analytics internship to apply skills in data analysis, dashboarding, predictive analytics, reporting, and data-driven decision making.

## EDUCATION

**Bachelor of Engineering in Artificial Intelligence and Data Science**

**Expected Graduation: 2027**

K. J. Somaiya School of Engineering

## TECHNICAL SKILLS

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|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b>                        | Python, SQL, C++, Java, JavaScript                                                                                                                                               |
| <b>Data Analytics &amp; Visualization:</b> | MS Excel, Power BI, Tableau, Pandas, Exploratory Data Analysis (EDA), Data Cleaning & Wrangling, Dashboarding, Geospatial Analysis (QGIS)                                        |
| <b>Databases &amp; ETL:</b>                | SQL, PostgreSQL, SQLite, SQLAlchemy 2.0, asyncpg, Relational Database Design, ER Modeling, Query Optimization, ETL Pipelines, Data Warehousing, OLAP/OLTP, Star/Snowflake Schema |
| <b>Machine Learning:</b>                   | scikit-learn, Regression, Classification, Clustering, Feature Engineering, Dimensionality Reduction (PCA)                                                                        |
| <b>Machine Learning Methods:</b>           | SVM, Decision Trees, k-NN, Naive Bayes, k-Means, DBSCAN                                                                                                                          |
| <b>Deep Learning &amp; AI:</b>             | TensorFlow, PyTorch, NLP, Transformer Architectures, LLM API Integration (Groq, Gemini), AI Agent Workflows, Permission-Gated Task Execution                                     |
| <b>Development &amp; Cloud:</b>            | AWS, FastAPI, Flask, Node.js, Express, React.js, TypeScript, Tailwind CSS, REST APIs, Asynchronous Python, Git, VSCode                                                           |
| <b>Security &amp; Testing:</b>             | Google OAuth 2.0, Firebase Admin SDK, JWT Authentication, HMAC-SHA256 Token Validation, Permission Scopes, Audit Logging, Async Pytest, Jest, Supertest                          |

## EXPERIENCE

**AI Research Intern**

Feb 2026 – May 2026

EllanorAI

- Contributed to reinforcement learning and alignment workflows, working with data pipelines supporting RLHF/RLAIF experimentation and model evaluation.
- Assisted in training, fine-tuning, and evaluating Large Language Models, analyzing model outputs and evaluation results across research workflows.
- Contributed to an in-house RL alignment library (ThinkRL), developing modular APIs, reward modeling components, and scalable training pipelines.
- Designed and prototyped transformer-based deep learning architectures for language-model research and experimentation.

**Web3 Intern**

Feb 2025 – Mar 2025

Blockwee

- Developed web platforms with Peachworld supporting live user engagement and large-scale content delivery.
- Implemented scalable backend APIs and contributed to real-time data processing pipelines for application workflows.
- Collaborated across design, product, and engineering teams to deliver full-stack features end-to-end.

## PROJECTS

**Crewmate — Gamified AI Co-Founder Console**

Independent Project | GitHub

- Architected and built a full-stack, gamified AI co-founder platform (FastAPI, PostgreSQL, React 19 + TypeScript,

Tailwind v4) featuring a sleek console dashboard and real agent execution loops.

- Engineered a boundary-gated autonomy system where the AI organically levels up (Level 1 to 5), capable of transforming single natural-language prompts into fully drafted Google Slides presentations, automated Gmail complaint resolutions, and dynamic Google Forms.
- Built an asynchronous execution engine using Groq LLM inference to orchestrate 10+ sandboxed business tools, maintaining an immutable, append-only mission audit log via SQLAlchemy 2.0 to securely track all AI operations.
- Implemented secure, stateless dual-layer authentication using Firebase Admin SDK and Google OAuth 2.0, utilizing HMAC-SHA256 signatures for robust token validation without server-side sessions; covered by an asynchronous Pytest suite.

### **Nexus CRM — Full-Stack Support Ticketing Platform**

Independent Project | GitHub

- Architected and built a full-stack, multi-tenant support-ticketing CRM (Node.js/Express, SQLite via better-sqlite3, React 19 + TypeScript) with ticket lifecycle tracking, filtering/search, and Kanban workflows.
- Designed a normalized relational schema (organizations, users, tickets, notes) with foreign-key cascades, composite indexes, and an atomic per-tenant ticket-ID counter to prevent race conditions.
- Built a REST analytics API computing status distribution, average resolution time, and 14-day created-vs-closed trends, visualized through an interactive dashboard (Recharts).
- Implemented dual-layer authentication (JWT + Firebase Admin SDK for Google OAuth) with bcrypt hashing, rate limiting, and a validated, versioned REST API; covered with an 11-test Jest/Supertest suite.

### **Soil Health Data Extraction & Analytics Pipeline**

Independent Project | GitHub

- Reverse-engineered a government geospatial portal (soilhealth.dac.gov.in) and built an end-to-end Python pipeline to extract structured data from a map-based interface not exposed through standard API/HTML endpoints.
- Designed a 5-stage data workflow covering metadata resolution, tile generation, image-based marker detection using Pillow, coordinate-to-feature lookup through WMS GetFeatureInfo, and structured CSV export.
- Transformed visual map data into a clean, traceable dataset with source lineage across 17 fields per record, enabling structured downstream analysis.
- Built a parameterized command-line interface using argparse with filtering by state, district, cycle, and nutrient for reproducible data extraction and analysis.
- Validated extracted records against live portal data for Maharashtra → Satara, 2025–26 cycle, verifying the accuracy of the resulting dataset against real nutrient records.

### **CivicIQ — AI-Powered Election Education Assistant**

Hack2skill PromptWars Hackathon | GitHub

- Built an election-education assistant using the Google Gemini API and Google Cloud services, integrating AI capabilities into a Flask + React application.
- Developed the application as a data-driven platform for delivering structured, context-aware information to users.
- Authored the project SOP connecting the solution to UN SDG 4 (Quality Education) for the FutureMinds Summit, Thailand.

### **SABER College Chatbot**

SIH Hackathon 2024

- Architected a campus-scale chatbot using LLaMA and Flask, providing context-aware support for a large user base.
- Led end-to-end feature development and collaborated within a cross-functional team to build and deploy scalable application components.

## **CERTIFICATIONS**

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Virtual PromptWars by Google for Developers (Hack2skill) | Digital Transformation (University of Virginia) | Leadership Skills (IIM Ahmedabad) | Financial Markets (Yale University) | Introduction to Large Language Models (Google Cloud) | Introduction to Generative AI (Google Cloud) | Introduction to Responsible AI (Google Cloud)

## **ADDITIONAL QUALIFICATIONS**

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- Finalist in 3 case study competitions: CaseQuest 2026 (KJSSE E-Summit), MergeMania 2026 (SPIT E-Summit), and FCRIT Green Club Ideathon 2026.
- Experienced in presenting technical work and results to peers and senior stakeholders in multicultural, cross-functional environments.